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EDUCACIÓN

SECRETARÍA DE EDUCACIÓN, CIENCIA, TECNOLOGÍA E INNOVACIÓN

TESVG
VILLA GUERRERO



TECNOLÓGICO
NACIONAL DE MÉXICO



2026. "Año del Humanismo Mexicano en el Estado de México"

Tecnológico de estudios Superiores de Villa Guerrero.

Ingeniería en Sistemas Computacionales

UNIDAD DE APRENDIZAJE:

SISTEMAS OPERATIVOS

PRÁCTICA:

PRÁCTICA 1. "EXPLORANDO PROCESOS CON PS EN LINUX"

PRESENTA:

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3501

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Introducción

En esta práctica se explorará el manejo de procesos en un sistema operativo Linux mediante el uso del comando ps. Este comando permite consultar los procesos que se encuentran activos en el sistema y obtener información relacionada con ellos. A lo largo de la práctica se utilizarán diferentes opciones de ps para visualizar los procesos del usuario actual, mostrar información detallada, filtrar procesos específicos y consultar los procesos asociados a un determinado usuario mediante su ID. Asimismo, se obtendrá información específica de cada proceso, como su identificador de proceso (PID) y el nombre del comando que se está ejecutando.

El uso de estas opciones permitirá conocer diferentes formas de consultar y organizar la información proporcionada por el sistema sobre los procesos activos. Finalmente, se utilizará ps para ordenar los procesos de acuerdo con su porcentaje de uso del CPU, lo que permitirá observar cuáles son los procesos que presentan un mayor consumo de este recurso. De esta manera, la práctica permitirá familiarizarse con el uso de la terminal de Linux y con la información que proporciona el sistema para la identificación y consulta de sus procesos.

Objetivo

Aprender a utilizar el comando ps para visualizar, filtrar y comprender los procesos activos en un sistema Linux.

Requisitos

1. Sistema Linux (Debian, Ubuntu, Arch, etc.)
2. Terminal activa



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1. Ver procesos del usuario actual (ps)

Muestra los procesos que pertenecen al usuario que ejecutó la terminal y que están asociados con la sesión actual. En la información obtenida se puede observar el PID, la terminal utilizada, el tiempo de ejecución y el nombre del comando correspondiente a cada proceso.

```
user@debian:~$ ps
  PID TTY          TIME CMD
 3801 pts/0    00:00:00 bash
 3805 pts/0    00:00:00 ps
user@debian:~$
```

2. Ver procesos con formato extendido (ps -ef)

Muestra de manera detallada los procesos que se están ejecutando en el sistema. La información incluye datos como el usuario que ejecuta cada proceso, su PID, el PPID o proceso padre, la hora de inicio, el tiempo de ejecución y el comando utilizado.

```
Oct 4 3:11 PM
user@debian:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1        0  0 14:57 ?        00:00:01 /sbin/init components splash
root         2        0  0 14:57 ?        00:00:00 [kthreadd]
root         3        2  0 14:57 ?        00:00:00 [pool.workerqueue.release]
root         4        2  0 14:57 ?        00:00:00 [kworker/R-kvfree_rcu_reclai
root         5        2  0 14:57 ?        00:00:00 [kworker/R-rcu_gp]
root         6        2  0 14:57 ?        00:00:00 [kworker/R-sync_wq]
root         7        2  0 14:57 ?        00:00:00 [kworker/R-slub_flushwq]
root         8        2  0 14:57 ?        00:00:00 [kworker/R-netns]
root        10        2  0 14:57 ?        00:00:00 [kworker/R-0H-events_highpri
root        11        2  0 14:57 ?        00:00:00 [kworker/0:1-events]
root        13        2  0 14:57 ?        00:00:00 [kworker/R-mm_percpu_wq]
root        14        2  0 14:57 ?        00:00:00 [rcu_tasks_kthread]
root        15        2  0 14:57 ?        00:00:00 [rcu_tasks_rude_kthread]
root        16        2  0 14:57 ?        00:00:00 [rcu_tasks_trace_kthread]
root        17        2  0 14:57 ?        00:00:00 [ksftirqd/0]
root        18        2  0 14:57 ?        00:00:00 [rcu_preempt]
root        19        2  0 14:57 ?        00:00:00 [rcu_exp_par_gp_kthread work
root        20        2  0 14:57 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root        21        2  0 14:57 ?        00:00:00 [migration/0]
root        22        2  0 14:57 ?        00:00:00 [idle_inject/0]
root        23        2  0 14:57 ?        00:00:00 [cpuhp/8]
root        24        2  0 14:57 ?        00:00:00 [cpuhp/2]
root        25        2  0 14:57 ?        00:00:00 [idle_inject/2]
root        26        2  0 14:57 ?        00:00:00 [migration/2]
root        27        2  0 14:57 ?        00:00:00 [ksftirqd/2]
root        29        2  0 14:57 ?        00:00:00 [kworker/2:0H-events_highpri
root        30        2  0 14:57 ?        00:00:00 [cpuhp/4]
root        31        2  0 14:57 ?        00:00:00 [idle_inject/4]
root        32        2  0 14:57 ?        00:00:00 [migration/4]
root        33        2  0 14:57 ?        00:00:00 [ksftirqd/4]
root        35        2  0 14:57 ?        00:00:00 [kworker/4:0H-events_highpri
root        36        2  0 14:57 ?        00:00:00 [cpuhp/6]
root        37        2  0 14:57 ?        00:00:00 [idle_inject/6]
root        38        2  0 14:57 ?        00:00:00 [migration/6]
root        39        2  0 14:57 ?        00:00:00 [ksftirqd/6]
root        40        2  0 14:57 ?        00:00:00 [kworker/6:0-events]
root        41        2  0 14:57 ?        00:00:00 [kworker/6:0H-events_highpri
root        42        2  0 14:57 ?        00:00:00 [cpuhp/8]
root        43        2  0 14:57 ?        00:00:00 [idle_inject/8]
root        44        2  0 14:57 ?        00:00:00 [migration/8]
root        45        2  0 14:57 ?        00:00:00 [ksftirqd/8]
root        47        2  0 14:57 ?        00:00:00 [kworker/8:0H-events_highpri
root        48        2  0 14:57 ?        00:00:00 [cpuhp/10]
root        49        2  0 14:57 ?        00:00:00 [idle_inject/10]
root        50        2  0 14:57 ?        00:00:00 [migration/10]
root        51        2  0 14:57 ?        00:00:00 [ksftirqd/10]
root        53        2  0 14:57 ?        00:00:00 [kworker/10:0H-events_highpr
root        54        2  0 14:57 ?        00:00:00 [cpuhp/12]
root        55        2  0 14:57 ?        00:00:00 [idle_inject/12]
root        56        2  0 14:57 ?        00:00:00 [migration/12]
root        57        2  0 14:57 ?        00:00:00 [ksftirqd/12]
root        58        2  0 14:57 ?        00:00:00 [kworker/12:0-events]
root        59        2  0 14:57 ?        00:00:00 [kworker/12:0H-events_highpr
root        60        2  0 14:57 ?        00:00:00 [cpuhp/13]
root        61        2  0 14:57 ?        00:00:00 [idle_inject/13]
root        62        2  0 14:57 ?        00:00:00 [migration/13]
root        63        2  0 14:57 ?        00:00:00 [ksftirqd/13]
root        65        2  0 14:57 ?        00:00:00 [kworker/13:0H-events_highpr
root        66        2  0 14:57 ?        00:00:00 [cpuhp/14]
root        67        2  0 14:57 ?        00:00:00 [idle_inject/14]
```



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Oct 4 3:12 PM				
user@debian:~				
root	68	2	0 14:57 ?	00:00:00 [migration/14]
root	69	2	0 14:57 ?	00:00:00 [ksoftirqd/14]
root	70	2	0 14:57 ?	00:00:00 [kworker/14:0-events]
root	71	2	0 14:57 ?	00:00:00 [kworker/14:0H-events_highpr
root	72	2	0 14:57 ?	00:00:00 [cpuhp/15]
root	73	2	0 14:57 ?	00:00:00 [idle_inject/15]
root	74	2	0 14:57 ?	00:00:00 [migration/15]
root	75	2	0 14:57 ?	00:00:00 [ksoftirqd/15]
root	77	2	0 14:57 ?	00:00:00 [kworker/15:0H-events_highpr
root	78	2	0 14:57 ?	00:00:00 [cpuhp/1]
root	79	2	0 14:57 ?	00:00:00 [idle_inject/1]
root	80	2	0 14:57 ?	00:00:00 [migration/1]
root	81	2	0 14:57 ?	00:00:00 [ksoftirqd/1]
root	83	2	0 14:57 ?	00:00:00 [kworker/1:0H-events_highpri
root	84	2	0 14:57 ?	00:00:00 [cpuhp/3]
root	85	2	0 14:57 ?	00:00:00 [idle_inject/3]
root	86	2	0 14:57 ?	00:00:00 [migration/3]
root	87	2	0 14:57 ?	00:00:00 [ksoftirqd/3]
root	89	2	0 14:57 ?	00:00:00 [kworker/3:0H-events_highpri
root	90	2	0 14:57 ?	00:00:00 [cpuhp/5]
root	91	2	0 14:57 ?	00:00:00 [idle_inject/5]
root	92	2	0 14:57 ?	00:00:00 [migration/5]
root	93	2	0 14:57 ?	00:00:00 [ksoftirqd/5]
root	95	2	0 14:57 ?	00:00:00 [kworker/5:0H-events_highpri
root	96	2	0 14:57 ?	00:00:00 [cpuhp/7]
root	97	2	0 14:57 ?	00:00:00 [idle_inject/7]
root	98	2	0 14:57 ?	00:00:00 [migration/7]
root	99	2	0 14:57 ?	00:00:00 [ksoftirqd/7]
root	101	2	0 14:57 ?	00:00:00 [kworker/7:0H-events_highpri
root	102	2	0 14:57 ?	00:00:00 [cpuhp/9]
root	103	2	0 14:57 ?	00:00:00 [idle_inject/9]
root	104	2	0 14:57 ?	00:00:00 [migration/9]
root	105	2	0 14:57 ?	00:00:00 [ksoftirqd/9]
root	107	2	0 14:57 ?	00:00:00 [kworker/9:0H-events_highpri
root	108	2	0 14:57 ?	00:00:00 [cpuhp/11]
root	109	2	0 14:57 ?	00:00:00 [idle_inject/11]
root	110	2	0 14:57 ?	00:00:00 [migration/11]
root	111	2	0 14:57 ?	00:00:00 [ksoftirqd/11]
root	112	2	0 14:57 ?	00:00:00 [kworker/11:0-events]
root	113	2	0 14:57 ?	00:00:00 [kworker/11:0H-events_highpr
root	120	2	0 14:57 ?	00:00:00 [kdevtmpfs]
root	121	2	0 14:57 ?	00:00:00 [kworker/R-inet_frag_wq]
root	122	2	0 14:57 ?	00:00:00 [kauditd]
root	123	2	0 14:57 ?	00:00:00 [khungtaskd]
root	124	2	0 14:57 ?	00:00:00 [oom_reaper]
root	125	2	0 14:57 ?	00:00:00 [kworker/R-writeback]
root	126	2	0 14:57 ?	00:00:00 [kcompactd0]
root	127	2	0 14:57 ?	00:00:00 [ksmd]
root	128	2	0 14:57 ?	00:00:00 [khugepaged]
root	129	2	0 14:57 ?	00:00:00 [kworker/R-kintegrityd]
root	130	2	0 14:57 ?	00:00:00 [kworker/R-kblockd]
root	131	2	0 14:57 ?	00:00:00 [kworker/R-bioq_punt_bio]
root	132	2	0 14:57 ?	00:00:00 [kworker/12:1-mm_percpu_wq]
root	133	2	0 14:57 ?	00:00:00 [irq/9-acpi]
root	134	2	0 14:57 ?	00:00:00 [kworker/13:1-cgroup_free]
root	135	2	0 14:57 ?	00:00:00 [kworker/1:1-mm_percpu_wq]
root	136	2	0 14:57 ?	00:00:00 [kworker/2:1-events_freezabl
root	139	2	0 14:57 ?	00:00:00 [kworker/R-tpm_dev_wq]
root	140	2	0 14:57 ?	00:00:00 [kworker/R-edac_poller]
root	141	2	0 14:57 ?	00:00:00 [kworker/R-devfreq_wq]
root	142	2	0 14:57 ?	00:00:00 [kworker/R-quota_events_unbo
root	144	2	0 14:57 ?	00:00:00 [kworker/14:1-events]

Oct 4 3:12 PM				
user@debian:~				
root	145	2	0 14:57 ?	00:00:00 [kworker/14:1H-events_highpr
root	146	2	0 14:57 ?	00:00:00 [kswapd0]
root	154	2	0 14:57 ?	00:00:00 [kworker/R-kthrotld]
root	158	2	0 14:57 ?	00:00:00 [irq/122-aerdrv]
root	159	2	0 14:57 ?	00:00:00 [irq/123-aerdrv]
root	160	2	0 14:57 ?	00:00:00 [irq/123-pciehp]
root	161	2	0 14:57 ?	00:00:00 [irq/124-aerdrv]
root	162	2	0 14:57 ?	00:00:00 [irq/125-aerdrv]
root	163	2	0 14:57 ?	00:00:00 [irq/127-pciehp]
root	164	2	0 14:57 ?	00:00:00 [irq/129-pciehp]
root	165	2	0 14:57 ?	00:00:00 [kworker/R-acpi_thermal_pm]
root	166	2	0 14:57 ?	00:00:00 [hung]
root	167	2	0 14:57 ?	00:00:00 [kworker/R-hfi-updates]
root	168	2	0 14:57 ?	00:00:00 [kworker/R-mlid]
root	169	2	0 14:57 ?	00:00:00 [kworker/R-lpss_addrconf]
root	170	2	0 14:57 ?	00:00:00 [kworker/12:1H-events_highpr
root	175	2	0 14:57 ?	00:00:00 [kworker/R-kstrp]
root	177	2	0 14:57 ?	00:00:02 [kworker/u65:0-1915_flip]
root	243	2	0 14:57 ?	00:00:00 [kworker/6:1H-events_highpri
root	266	2	0 14:57 ?	00:00:00 [kworker/13:1H-events_highpr
root	275	2	0 14:57 ?	00:00:00 [kworker/2:1H-events_highpri
root	290	2	0 14:57 ?	00:00:00 [kworker/4:1H-kblockd]
root	291	2	0 14:57 ?	00:00:00 [kworker/3:2-events]
root	292	2	0 14:57 ?	00:00:00 [kworker/9:1H-events_highpri
root	327	2	0 14:57 ?	00:00:00 [kworker/6:2-events]
root	330	2	0 14:57 ?	00:00:00 [kworker/11:1H-events_highpr
root	331	2	0 14:57 ?	00:00:00 [kworker/6:1H-events_highpri
root	332	2	0 14:57 ?	00:00:00 [kworker/1:1H-events_highpri
root	333	2	0 14:57 ?	00:00:00 [kworker/15:1H-events_highpr
root	334	2	0 14:57 ?	00:00:00 [kworker/7:1H-events_highpri
root	335	2	0 14:57 ?	00:00:00 [kworker/2:1H-kblockd]
root	336	2	0 14:57 ?	00:00:00 [kworker/8:1H-events_highpri
root	337	2	0 14:57 ?	00:00:00 [kworker/5:1H-kblockd]
root	338	2	0 14:57 ?	00:00:00 [kworker/10:1H-events_highpr
root	340	2	0 14:57 ?	00:00:00 [kworker/5:1-mm_percpu_wq]
root	343	2	0 14:57 ?	00:00:00 [wq_chngd]
root	347	2	0 14:57 ?	00:00:00 [irq/166-ITES570:00]
root	348	2	0 14:57 ?	00:00:06 [irq/183-ASCE1200:00]
root	350	2	0 14:57 ?	00:00:00 [irq/184-aerdrv]
root	351	2	0 14:57 ?	00:00:00 [kworker/5:2-events]
root	352	2	0 14:57 ?	00:00:00 [kworker/R-nvme-wq]
root	353	2	0 14:57 ?	00:00:00 [kworker/R-nvme-reset-wq]
root	354	2	0 14:57 ?	00:00:00 [kworker/R-nvme-delete-wq]
root	355	2	0 14:57 ?	00:00:00 [kworker/R-nvme-auth-wq]
root	356	2	0 14:57 ?	00:00:00 [kworker/6:2-1915-unordered]
root	357	2	0 14:57 ?	00:00:00 [kworker/9:1-mm_percpu_wq]
root	358	2	0 14:57 ?	00:00:00 [kworker/R-tnn]
root	359	2	0 14:57 ?	00:00:00 [card0-crt0]
root	360	2	0 14:57 ?	00:00:00 [card0-crt1]
root	361	2	0 14:57 ?	00:00:00 [card0-crt2]
root	362	2	0 14:57 ?	00:00:00 [card0-crt3]
root	363	2	0 14:57 ?	00:00:00 [kworker/7:1-1915-unordered]
root	366	2	0 14:57 ?	00:00:00 [kworker/10:1-mm_percpu_wq]
root	367	2	0 14:57 ?	00:00:00 [kworker/11:2-events]
root	368	2	0 14:57 ?	00:00:00 [kworker/R-USB000-00-con1]
root	369	2	0 14:57 ?	00:00:00 [scsi eh 0]
root	370	2	0 14:57 ?	00:00:00 [kworker/R-scsi_tmf_0]
root	371	2	0 14:57 ?	00:00:00 [usb-storage]
root	372	2	0 14:57 ?	00:00:00 [kworker/R-usb]
root	381	2	0 14:57 ?	00:00:00 [kworker/R-md]
root	382	2	0 14:57 ?	00:00:00 [kworker/R-md_bitmap]
root	388	2	0 14:57 ?	00:00:00 [kworker/R-raid5wq]



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```
Oct 4 3:13 PM
user@debian:~$ cat /etc/passwd

root:445:2:0:14:57:?:0:0:0:0:[worker/10:2-cgroup_free]
root:573:2:0:14:57:?:0:0:0:0:[psimem]
root:587:1:0:14:57:?:0:0:0:0:/usr/lib/systemd/systemd-jou
root:645:1:0:14:57:?:0:0:0:0:/usr/lib/systemd/systemd-ude
root:647:2:0:14:57:?:0:0:0:0:[psimem]
root:737:2:0:14:57:?:0:0:0:0:[irq/203-mei_ne]
root:738:2:0:14:57:?:0:0:0:0:[worker/8:2-1915-unordered]
root:742:2:0:14:57:?:0:0:0:0:[worker/4:3-events]
root:743:2:0:14:57:?:0:0:0:0:[worker/R-cryptd]
root:744:2:0:14:57:?:0:0:0:0:[worker/R-led_workqueue]
root:749:2:0:14:57:?:0:0:0:0:[worker/R-cfg80211]
root:750:2:0:14:57:?:0:0:0:0:[irq/16-processor_thermal_de
root:768:2:0:14:57:?:0:0:0:0:[worker/u64:6-1915]
root:778:2:0:14:57:?:0:0:0:0:[worker/u64:11-usb]
root:784:2:0:14:57:?:0:0:0:0:[worker/u64:14-events_unbou
root:785:2:0:14:57:?:0:0:0:0:[worker/7:2-cgroup_release]
root:839:2:0:14:57:?:0:0:0:0:[worker/13:2-events]
root:847:2:0:14:57:?:0:0:0:0:[worker/15:2-mm_percpu_wq]
root:855:2:0:14:57:?:0:0:0:0:[worker/u65:14-1915_flip]
root:979:2:0:14:57:?:0:0:0:0:[irq/205-twd9-pci]
root:1535:1:0:14:57:?:0:0:0:0:/usr/libexec/accounts-daemon
avahi:1536:1:0:14:57:?:0:0:0:0:avahi-daemon:running [debia
root:1537:1:0:14:57:?:0:0:0:0:/usr/libexec/bluetooth/bluet
root:1539:1:0:14:57:?:0:0:0:0:/usr/sbin/cron -f
message+ 1541:1:0:14:57:?:0:0:0:0:/usr/bin/dbus-daemon --syste
root:1544:1:0:14:57:?:0:0:0:0:/usr/libexec/low-memory-monit
root:1545:1:0:14:57:?:0:0:0:0:/usr/lib/polkit-1/polkitd --
root:1546:1:0:14:57:?:0:0:0:0:/usr/sbin/smartd -n
root:1547:1:0:14:57:?:0:0:0:0:/usr/libexec/switcheroo-cont
root:1549:1:0:14:57:?:0:0:0:0:/usr/lib/systemd/systemd-log
root:1550:1:0:14:57:?:0:0:0:0:/usr/libexec/udisks2/udisksd
avahi:1551:1536:0:14:57:?:0:0:0:0:avahi-daemon:chroot helper
root:1594:1:0:14:57:?:0:0:0:0:/usr/sbin/NetworkManager --n
root:1595:1:0:14:57:?:0:0:0:0:/usr/sbin/wpa_supplicant -u
root:1603:2:0:14:57:?:0:0:0:0:[psimem]
root:1629:1:0:14:57:?:0:0:0:0:/usr/sbin/ModemManager
root:1630:1:0:14:57:?:0:0:0:0:/usr/libexec/bolt
root:1646:2:0:14:57:?:0:0:0:0:[krconmd]
root:1662:1:0:14:57:?:0:0:0:0:/usr/sbin/cupsd -l
root:1675:1:0:14:57:?:0:0:0:0:/usr/sbin/gdm3
root:1682:1675:0:14:57:?:0:0:0:0:gdm-session-worker [pan/gdm
user:1690:1:0:14:57:?:0:0:0:0:/usr/lib/systemd/systemd --u
user:1692:1690:0:14:57:?:0:0:0:0:[sd-pam]
user:1712:1690:0:14:57:?:0:0:0:0:/usr/bin/dbus-daemon --sessi
user:1714:1690:0:14:57:?:0:0:0:0:/usr/bin/pipewire
user:1716:1690:0:14:57:?:0:0:0:0:/usr/bin/pipewire -c filter-
user:1717:1690:0:14:57:?:0:0:0:0:/usr/bin/pipewire-lmber
user:1718:1690:0:14:57:?:0:0:0:0:/usr/bin/pipewire-pulse
user:1720:1690:0:14:57:?:0:0:0:0:/usr/bin/gnome-keyring-daemo
user:1721:1690:0:14:57:?:0:0:0:0:/usr/bin/npis-proxy
root:1730:1:0:14:57:?:0:0:0:0:/usr/libexec/rtkit-daemon
user:1734:1682:0:14:57:?:0:0:0:0:/usr/libexec/gdm-wayland-ses
user:1738:1734:0:14:57:?:0:0:0:0:/usr/libexec/gnome-session-b
user:1792:1690:0:14:57:?:0:0:0:0:/usr/libexec/gcr-ssh-agent -
user:1793:1690:0:14:57:?:0:0:0:0:/usr/libexec/gnome-session-c
user:1794:1690:0:14:57:?:0:0:0:0:/usr/bin/ssh-agent -D
user:1804:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfsd
user:1812:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfsd-fuse /run
user:1815:1690:0:14:57:?:0:0:0:0:/usr/libexec/gnome-session-b
user:1850:1815:0:14:57:?:0:0:0:0:/usr/libexec/at-spi-bus-lau
root:1851:1690:0:14:57:?:0:0:0:0:/usr/bin/gnome-shell
user:1869:1850:0:14:57:?:0:0:0:0:/usr/bin/dbus-daemon --confi
```

```
Oct 4 3:13 PM
user@debian:~$ cat /etc/passwd

user:1892:1690:0:14:57:?:0:0:0:0:/usr/libexec/gdm3
user:1894:1690:0:14:57:?:0:0:0:0:/usr/libexec/gdm3-permission
user:1919:1690:0:14:57:?:0:0:0:0:/usr/libexec/at-spi2-registr
colorD 1920:1:0:14:57:?:0:0:0:0:/usr/libexec/colorD
user:1954:1690:0:14:57:?:0:0:0:0:/usr/libexec/gnome-shell-cal
user:1962:1690:0:14:57:?:0:0:0:0:/usr/libexec/dconf-service
user:1964:1690:0:14:57:?:0:0:0:0:/usr/bin/gjs -m /usr/share/g
user:1985:1690:0:14:57:?:0:0:0:0:/usr/bin/dbus-daemon --panel
root:1986:1:0:14:57:?:0:0:0:0:/usr/libexec/upowerd
user:1987:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-ally-settin
user:1989:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-color
user:1990:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-datetime
user:1991:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-housekeepin
user:1994:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-keyboard
user:1998:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-media-keys
user:1999:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-power
user:2000:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-print-notif
user:2005:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-rfkill
user:2006:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-screensaver
user:2008:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-sharing
user:2009:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-smartcard
user:2013:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-sound
user:2014:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-usb-protect
user:2015:1690:0:14:57:?:0:0:0:0:/usr/libexec/gsd-vacom
user:2017:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-evolution-sourc
user:2018:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-udisks2-vo
user:2070:1690:0:14:57:?:0:0:0:0:/usr/bin/gjs -m /usr/share/g
user:2115:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-mtp-volume
user:2143:1815:0:14:57:?:0:0:0:0:/usr/libexec/gsd-disk-utiliti
user:2151:1815:1:14:57:?:0:0:0:0:/usr/bin/gnome-software --ga
user:2154:1815:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-evolution-data
user:2218:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-gphoto2-vo
user:2224:1690:0:14:57:?:0:0:0:0:/usr/libexec/gvfs-goa-volume
user:2229:1690:0:14:57:?:0:0:0:0:/usr/libexec/goa-daemon
user:2238:1690:0:14:58:?:0:0:0:0:/usr/libexec/goa-identity-se
user:2240:1690:0:14:58:?:0:0:0:0:/usr/libexec/gvfs-atc-volume
user:2254:1851:0:14:58:?:0:0:0:0:/usr/bin/wayland-0-rootle
user:2302:1690:0:14:58:?:0:0:0:0:/usr/libexec/gsd-printer
user:2325:1690:0:14:58:?:0:0:0:0:/usr/libexec/xdg-desktop-por
user:2329:1690:0:14:58:?:0:0:0:0:/usr/libexec/gvfs-evolution-calen
user:2333:1690:0:14:58:?:0:0:0:0:/usr/libexec/xdg-document-po
root:2342:2333:0:14:58:?:0:0:0:0:/usr/bin/fusemount3 -o rw,nosuid,nod
user:2359:1690:0:14:58:?:0:0:0:0:/usr/libexec/gsd-xsettings
user:2355:1690:0:14:58:?:0:0:0:0:/usr/libexec/gvfs-evolution-adre
user:2387:1690:0:14:58:?:0:0:0:0:/usr/libexec/gvfsd-metadata
user:2409:1851:0:14:58:?:0:0:0:0:/usr/libexec/mutter-x11-fran
root:2446:1:0:14:58:?:0:0:0:0:/usr/sbin/cups-browsed
user:2459:1985:0:14:58:?:0:0:0:0:/usr/libexec/ibus-dconf
user:2460:1985:0:14:58:?:0:0:0:0:/usr/libexec/ibus-extension-
user:2467:1690:0:14:58:?:0:0:0:0:/usr/libexec/ibus-portal
user:2469:1690:0:14:58:?:0:0:0:0:/usr/libexec/localsearch-3
user:2470:1690:0:14:58:?:0:0:0:0:/usr/libexec/xdg-desktop-por
user:2483:1985:0:14:58:?:0:0:0:0:/usr/libexec/ibus-english-sin
user:2500:1690:0:14:58:?:0:0:0:0:/usr/libexec/ibus-x11
user:2515:1690:0:14:58:?:0:0:0:0:/usr/libexec/xdg-desktop-por
root:2542:1:0:14:58:?:0:0:0:0:/usr/libexec/wayd/wayd
root:2622:1:0:14:58:?:0:0:0:0:/usr/libexec/power-profiles-
user:2692:1851:0:14:58:?:0:0:1:56:/usr/lib/firefox-esr/firefox
user:2700:1690:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/crashr
user:2773:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2836:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2844:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2878:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2878:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2981:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:2990:2692:23:14:58:?:0:0:2:50:/usr/lib/firefox-esr/firefox
user:3004:2692:1:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:3127:2692:0:14:58:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:3611:1690:0:15:00:?:0:0:0:0:/usr/bin/gnome-calendar --ga
user:3668:1804:0:15:00:?:0:0:0:0:/usr/libexec/gvfsd-trash --s
root:4155:2:0:15:02:?:0:0:0:0:[worker/u64:8-events_unbou
root:4160:2:0:15:03:?:0:0:0:0:[worker/4:0-events]
root:4161:2:0:15:03:?:0:0:0:0:[worker/2:0-events]
root:4162:2:0:15:03:?:0:0:0:0:[worker/15:0]
root:4166:2:0:15:03:?:0:0:0:0:[worker/9:2]
root:4169:2:0:15:04:?:0:0:0:0:[worker/7:1-mm_percpu_wq]
user:4176:2692:0:15:04:?:0:0:0:0:/usr/lib/firefox-esr/firefox
user:4284:2692:0:15:04:?:0:0:0:0:/usr/lib/firefox-esr/firefox
root:4316:2:0:15:04:?:0:0:0:0:[worker/8:1-mm_percpu_wq]
root:4343:2:0:15:05:?:0:0:0:0:[worker/1:0]
root:4539:2:0:15:06:?:0:0:0:0:[worker/14:2-events_power.e
root:4551:2:0:15:06:?:0:0:0:0:[worker/11:1]
root:4577:2:0:15:06:?:0:0:0:0:[worker/10:0-events]
root:5201:2:0:15:08:?:0:0:0:0:[worker/u65:2]
user:5561:1690:1:15:08:?:0:0:0:0:/usr/libexec/gnome-terminal-
user:5568:5561:0:15:08:?:0:0:0:0:/usr/bin/bash
root:5650:2:0:15:09:?:0:0:0:0:[worker/13:0-mm_percpu_wq]
user:5674:2692:0:15:09:?:0:0:0:0:/usr/lib/firefox-esr/firefox
root:5716:2:0:15:10:?:0:0:0:0:[worker/4:1-1915-unordered]
root:5725:2:0:15:10:?:0:0:0:0:[worker/7:0-cgroup_free]
root:5731:2:0:15:10:?:0:0:0:0:[worker/2:2-events]
user:5747:5568:0:15:10:?:0:0:0:0:/usr/bin/ps -ef
```




2026. "Año del Humanismo Mexicano en el Estado de México"

3. Filtrar procesos (ps -ef | grep Firefox)

Utiliza grep para localizar dentro de la lista generada por ps -ef únicamente las líneas que contienen la palabra firefox. De esta manera se pueden identificar fácilmente los procesos relacionados con el navegador Firefox sin tener que revisar toda la lista.

```
user@debian:~$ ps -ef | grep firefox
user      2692   1511 13 14:58 ?        00:02:22 /usr/lib/firefox-esr/firefox-esr
user      2700   1690 0 14:58 ?        00:00:00 /usr/lib/firefox-esr/crashhelper 2692 9 /tmp/ 11
user      2773   2692 0 14:58 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefHandle 0:26640 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
chHandle 4 -initialChannelId {edf98ab5-af07-455d-be33-bd434803f59f} -parentPid 2692 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 1 socket
user      2836   2692 0 14:58 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:28015 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {0daa1664-345d-431c-9af7-2bc0a8f2c22} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 2 tab
user      2844   2692 0 14:58 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefHandle 0:28060 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
chHandle 4 -initialChannelId {4b4077cf-211c-492b-a6a1-a7c2907605f1} -parentPid 2692 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 3 rdd
user      2878   2692 0 14:58 ?        00:00:02 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:28266 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {97d3c21f-ca38-4aec-b94c-79b1e650067e} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 4 tab
user      2981   2692 0 14:58 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -sandboxingKind 0 -prefHandle 0:39629 -prefMapHandle 1:268799 -sandboxReporter 2 -
chrootClient 3 -ipHandle 4 -initialChannelId {cf6289a5-dee3-4efa-9cf9-953cb99ffac} -parentPid 2692 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 6 utility
user      2998   2692 22 14:58 ?        00:03:47 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:37796 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {f3977eal-2cdd-4c78-922f-3cab4d692806} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 8 tab
user      3004   2692 0 14:58 ?        00:00:09 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:37796 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {2b1741da-3cf6-45f9-ac46-d0b97e4c1515} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 9 tab
user      3127   2692 0 14:59 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:38726 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {17f7bd96-cbae-44be-acf5-6027eaf26221} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 10 tab
user      4284   2692 0 15:04 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:42597 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {49610960-1ec2-43fc-9d62-58f3bc07b9a9} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 19 tab
user      5674   2692 0 15:09 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:42597 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {1ae13954-0d38-44b1-8c7a-9524a3a252c5} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 20 tab
user      6022   2692 0 15:14 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefHandle 0:42597 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipHandle 5 -initialChannelId {30ac0285-87f6-4f6d-ad12-d8859715a327} -parentPid 2692 -crashReporter 6 -crashHelper 7 -greonni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 21 tab
user      6089   5568 0 15:15 pts/0    00:00:00 grep firefox
user@debian:~$
```

4. Ver procesos por ID de usuario (ps -u nombre_usuario)

Permite mostrar los procesos que pertenecen a un usuario específico. En esta práctica se utilizó user en lugar de nombre_usuario, ya que user es el nombre de la cuenta utilizada en el sistema Debian. La salida muestra los procesos asociados a dicha cuenta, incluyendo su PID, terminal, tiempo de ejecución y nombre del comando.

```
Oct 4 3:18 PM
user@debian:~$ ps -u user
PID TTY      TIME CMD
1690 ?        00:00:00 systemd
1692 ?        00:00:00 (sd-pan)
1712 ?        00:00:00 dbus-daemon
1714 ?        00:00:00 pipewire
1716 ?        00:00:00 pipewire
1717 ?        00:00:00 wireplumber
1718 ?        00:00:00 pipewire-pulse
1720 ?        00:00:00 gnome-keyring-d
1721 ?        00:00:00 mpris-proxy
1734 tty2    00:00:00 gdm-wayland-ses
1738 tty2    00:00:00 gnome-session-b
1792 ?        00:00:00 gcr-sh-agent
1793 ?        00:00:00 gnome-session-c
1794 ?        00:00:00 ssh-agent
1804 ?        00:00:00 gvfsd
1812 ?        00:00:00 gvfsd-fuse
1815 ?        00:00:00 gnome-session-b
1856 ?        00:00:00 at-spi-bus-laun
1851 ?        00:01:46 gnome-shell
1869 ?        00:00:00 dbus-daemon
1894 ?        00:00:00 xdg-permission-
1919 ?        00:00:00 at-spi2-registr
1954 ?        00:00:00 gnome-shell-cal
1962 ?        00:00:00 dconf-service
1964 ?        00:00:00 gjs
1985 ?        00:00:01 ibus-daemon
1987 ?        00:00:00 gsd-ally-settin
1989 ?        00:00:00 gsd-color
1990 ?        00:00:00 gsd-datetime
1991 ?        00:00:00 gsd-housekeepin
1994 ?        00:00:00 gsd-keyboard
1998 ?        00:00:00 gsd-media-keys
1999 ?        00:00:00 gsd-power
2000 ?        00:00:00 gsd-print-notif
2005 ?        00:00:00 gsd-rfkill
2006 ?        00:00:00 gsd-screensaver
2008 ?        00:00:00 gsd-sharing
2009 ?        00:00:00 gsd-smartcard
2013 ?        00:00:00 gsd-sound
2014 ?        00:00:00 gsd-usb-protect
2015 ?        00:00:00 gsd-wacom
2017 ?        00:00:00 evolution-sourc
2018 ?        00:00:00 gvfs-udisks2-vo
2070 ?        00:00:00 gjs
2115 ?        00:00:00 gvfs-mtp-volume
2143 ?        00:00:00 gsd-disk-utilit
2151 ?        00:00:09 gnome-software
2154 ?        00:00:00 evolution-alarm
2218 ?        00:00:00 gvfs-gphoto2-vo
2224 ?        00:00:00 gvfs-goa-volume
2229 ?        00:00:00 gsd-daemon
2238 ?        00:00:00 goa-identity-se
2240 ?        00:00:00 gvfs-afc-volume
2254 ?        00:00:00 xwayland
2302 ?        00:00:00 gsd-printer
2325 ?        00:00:00 xdg-desktop-por
2329 ?        00:00:00 evolution-calen
2333 ?        00:00:00 xdg-document-po
2359 ?        00:00:00 gsd-xsettings
2385 ?        00:00:00 evolution-adfse
```



2026. "Año del Humanismo Mexicano en el Estado de México"

```
2387 ? 00:00:00 gvfsd-metadata
2409 ? 00:00:00 mutter-x11-frag
2459 ? 00:00:00 ibus-dconf
2460 ? 00:00:01 ibus-extension-
2467 ? 00:00:00 ibus-portal
2469 ? 00:00:00 localsearch-3
2470 ? 00:00:00 xdg-desktop-por
2481 ? 00:00:00 ibus-engine-sim
2500 ? 00:00:00 ibus-x11
2515 ? 00:00:00 xdg-desktop-por
2592 ? 00:02:33 firefox-esr
2700 ? 00:00:00 crashhelper
2773 ? 00:00:00 Socket Process
2836 ? 00:00:00 WebExtensions
2844 ? 00:00:00 RDB Process
2878 ? 00:00:02 Privileged Cont
2981 ? 00:00:00 Utility Process
2990 ? 00:04:12 Isolated Web Co
3084 ? 00:00:10 Isolated Web Co
3127 ? 00:00:00 Isolated Web Co
3611 ? 00:00:00 gnome-calendar
3668 ? 00:00:00 gvfsd-trash
4284 ? 00:00:00 Web Content
5561 ? 00:00:22 gnome-terminal-
5568 pts/0 00:00:00 bash
5674 ? 00:00:00 Web Content
6022 ? 00:00:00 Web Content
6198 pts/0 00:00:00 ps
user@debian:~$
```

5. Ver solo PID y comando (ps -eo pid,comm)

Este comando muestra una lista de los procesos activos del sistema, indicando únicamente su PID, que es el identificador de cada proceso, y el nombre del comando que lo está ejecutando. En la imagen se observa la lista de procesos, donde cada uno aparece acompañado de su número de identificación, lo que permite reconocerlos y diferenciarlos fácilmente.

```
Oct 4 3:20 PM
user@debian:~$ ps -eo pid,comm
pid comm
1 system
2 kthreadd
3 pool_workqueue_release
4 kworker/R-kvfree_rcu_reclaim
5 kworker/R-rcu_gp
6 kworker/R-sync_wq
7 kworker/R-slab_flushwq
8 kworker/R-netns
10 kworker/0:0H-events_highpri
11 kworker/0:1-events
13 kworker/R-mm_percpu_wq
14 rcu_tasks_kthread
15 rcu_tasks_rude_kthread
16 rcu_tasks_trace_kthread
17 ksoftirqd/0
18 rcu_preempt
19 rcu_exp_par_gp_kthread_worker/0
20 rcu_exp_gp_kthread_worker
21 migration/0
22 idle_inject/0
23 cpuhp/0
24 cpuhp/2
25 idle_inject/2
26 migration/2
27 ksoftirqd/2
29 kworker/2:0H-events_highpri
30 cpuhp/4
31 idle_inject/4
32 migration/4
33 ksoftirqd/4
35 kworker/4:0H-events_highpri
36 cpuhp/6
37 idle_inject/6
38 migration/6
39 ksoftirqd/6
41 kworker/6:0H-events_highpri
42 cpuhp/8
43 idle_inject/8
44 migration/8
45 ksoftirqd/8
47 kworker/8:0H-events_highpri
48 cpuhp/10
49 idle_inject/10
50 migration/10
51 ksoftirqd/10
53 kworker/10:0H-events_highpri
54 cpuhp/12
55 idle_inject/12
56 migration/12
57 ksoftirqd/12
59 kworker/12:0H-events_highpri
60 cpuhp/14
61 idle_inject/14
62 migration/14
63 ksoftirqd/14
65 kworker/14:0H-events_highpri
66 cpuhp/16
67 idle_inject/16
68 migration/16
69 ksoftirqd/16
```

```
70 kworker/14:0-events
71 kworker/14:0H-events_highpri
72 cpuhp/15
73 idle_inject/15
74 migration/15
75 kssoftirqd/15
77 kworker/15:0H-events_highpri
78 cpuhp/1
79 idle_inject/1
80 migration/1
81 kssoftirqd/1
83 kworker/1:0H-events_highpri
84 cpuhp/3
85 idle_inject/3
86 migration/3
87 kssoftirqd/3
89 kworker/3:0H-events_highpri
90 cpuhp/5
91 idle_inject/5
92 migration/5
93 kssoftirqd/5
95 kworker/5:0H-events_highpri
96 cpuhp/7
97 idle_inject/7
98 migration/7
99 kssoftirqd/7
101 kworker/7:0H-events_highpri
102 cpuhp/9
103 idle_inject/9
104 migration/9
105 kssoftirqd/9
107 kworker/9:0H-events_highpri
108 cpuhp/11
109 idle_inject/11
110 migration/11
111 kssoftirqd/11
112 kworker/11:0-events
113 kworker/11:0H-events_highpri
120 kdevtmpfs
121 kworker/R-inet_frag_wq
122 kauditd
123 khungtaskd
124 oom_reaper
125 kworker/R-writeback
126 kcompactd0
127 ksmd
128 khugepaged
129 kworker/R-kintegrityd
130 kworker/R-kblockd
131 kworker/R-blkcg_punt_bio
132 kworker/12:1-mmc_percpu_wq
133 irq/9-acpi
134 kworker/13:1-cgroup_free
135 kworker/1:1-mm_percpu_wq
136 kworker/2:1-mm_percpu_wq
139 kworker/R-tpm_dev_wq
140 kworker/R-edac-poller
141 kworker/R-devfreq_wq
142 kworker/R-quota_events_unbound
145 kworker/14:1H-events_highpri
146 kswapd0
154 kworker/R-kthrotld
...

158 irq/122-aerdrv
159 irq/123-aerdrv
160 irq/123-pciehp
161 irq/124-aerdrv
162 irq/125-aerdrv
163 irq/127-pciehp
164 irq/129-pciehp
165 kworker/R-acpi_thermal_pm
166 hwrng
167 kworker/R-hfi_updates
168 kworker/R-mld
169 kworker/R-ipv6_addrconf
170 kworker/12:1H-events_highpri
175 kworker/R-krtmp
177 kworker/u65:0-1915_flip
243 kworker/0:1H-events_highpri
266 kworker/13:1H-events_highpri
275 kworker/2:1H-events_highpri
290 kworker/4:1H-events_highpri
292 kworker/9:1H-events_highpri
330 kworker/11:1H-events_highpri
331 kworker/6:1H-events_highpri
332 kworker/1:1H-events_highpri
333 kworker/15:1H-events_highpri
334 kworker/7:1H-events_highpri
335 kworker/3:1H-events_highpri
336 kworker/8:1H-events_highpri
337 kworker/5:1H-events_highpri
338 kworker/10:1H-events_highpri
340 kworker/5:1-mm_percpu_wq
343 watchdogd
347 irq/186-ITSS570:00
348 irq/183-ASCE1200:00
350 irq/184-aerdrv
352 kworker/R-nme-wq
353 kworker/R-nme-reset-wq
354 kworker/R-nme-delete-wq
355 kworker/R-nme-auth-wq
356 kworker/6:2-mm_percpu_wq
357 kworker/9:1-events
358 kworker/R-ttm
361 card0-crtcl
361 card0-crtc2
362 card0-crtc3
363 kworker/7:1-1915_unordered
366 kworker/10:1-mm_percpu_wq
368 kworker/R-USC000:00-con1
369 scsi_eh_0
370 kworker/R-scsi_tmf_0
371 usb-storage
372 kworker/R-uas
381 kworker/R-md
382 kworker/R-md_bitmap
388 kworker/R-raid5q
573 psimon
587 systemd-journal
645 systemd-udev
647 psimon
737 irq/203-mei_me
743 kworker/R-cryptd
744 kworker/R-led_workqueue
```




2026. "Año del Humanismo Mexicano en el Estado de México"

```
Oct 4 3:21 PM
user@debian:~$ ls -l /etc/passwd
-rw-r--r-- 1 root root 4096 Oct 4 3:21 /etc/passwd
749 kworker/u64:0-cgroup
750 irq/16-processor_thermal_device_pci
768 kworker/u64:6-events_unbound
784 kworker/u64:14-1915
847 kworker/13:2-mm_percpu_wq
855 kworker/u64:1-1915_flip
979 irq/265-rtw89_pci
1535 accounts-daemon
1536 avahi-daemon
1537 bluetoothd
1539 cron
1541 dbus-daemon
1544 low-memory-monit
1545 polkitd
1546 smartd
1547 switcheroo-cont
1549 systemd-logind
1550 udisksd
1551 avahi-daemon
1594 NetworkManager
1595 wpa_supplicant
1601 psimon
1629 ModemManager
1630 bolt
1646 krfcomd
1662 cupsd
1675 gdm3
1682 gdm-session-wor
1690 systemd
1692 (sd-pam)
1712 dbus-daemon
1714 pipewire
1716 pipewire
1717 wireplumber
1718 pipewire-pulse
1720 gnome-keyring-d
1721 nprism-proxy
1730 rtkit-daemon
1734 gdm-wayland-ses
1738 gnome-session-b
1792 gcr-ssh-agent
1793 gnome-session-c
1794 ssh-agent
1804 gvfsd
1812 gvfsd-fuse
1815 gnome-session-b
1850 at-spi-bus-lau
1851 gnome-shell
1869 dbus-daemon
1894 xdg-permission-
1919 at-spi2-registr
1920 colord
1954 gnome-shell-cal
1962 dconf-service
1964 gjs
1965 ibus-daemon
1966 upowerd
1987 gsd-ally-settin
1989 gsd-color
1990 gsd-datetime
1991 gsd-housekeepin
1994 gsd-keyboard

Oct 4 3:22 PM
user@debian:~$ ls -l /etc/passwd
-rw-r--r-- 1 root root 4096 Oct 4 3:22 /etc/passwd
1998 gsd-media-keys
1999 gsd-power
2000 gsd-print-notif
2005 gsd-rfkill
2006 gsd-screensaver
2008 gsd-sharing
2009 gsd-smartcard
2013 gsd-sound
2014 gsd-usb-protect
2015 gsd-wacom
2017 evolution-sourc
2018 gvfs-udisks2-vo
2070 gjs
2115 gvfs-mtp-volume
2143 gsd-disk-utilit
2151 gnome-software
2154 evolution-alarm
2218 gvfs-gphoto2-vo
2224 gvfs-goa-volume
2229 goa-daemon
2238 goa-identity-se
2240 gvfs-afc-volume
2254 Xwayland
2302 gsd-printer
2325 xdg-desktop-por
2329 evolution-calen
2333 xdg-document-po
2342 fuserroutin3
2359 gsd-xsettings
2385 evolution-adre
2387 gvfs-metadata
2409 mutter-x11-fram
2446 cups-browsed
2459 ibus-dconf
2460 ibus-extension-
2467 ibus-portal
2469 localSearch-3
2470 xdg-desktop-por
2481 ibus-engine-sim
2500 ibus-x11
2515 xdg-desktop-por
2542 fwupd
2622 power-profiles-
2692 firefox-esr
2700 crashhelper
2773 Socket Process
2836 WebExtensions
2844 RDP Process
2878 Privileged Cont
2981 Utility Process
2990 Isolated Web Co
3004 Isolated Web Co
3127 Isolated Web Co
3611 gnome-calendar
3668 gvfsd-trash
4155 kworker/u64:8-events_unbound
4168 kworker/4:8-events
4166 kworker/9:2
4169 kworker/3:1-events_freezable_pwr_efficient
4284 Web Content
4316 kworker/8:1-mm_percpu_wq
4343 kworker/1:0

4539 kworker/14:2-mm_percpu_wq
4551 kworker/11:1
4577 kworker/10:0-events
5561 gnome-terminal-
5568 bash
5650 kworker/13:0-mm_percpu_wq
5674 Web Content
5716 kworker/4:1-mm_percpu_wq
5725 kworker/7:0-cgroup_free
5731 kworker/2:2-events
5796 kworker/12:2-mm_percpu_wq
5804 kworker/3:0-mm_percpu_wq
5813 kworker/5:0-events
5828 kworker/0:0-events
5871 kworker/15:1-mm_percpu_wq
5915 kworker/8:0-events
6022 Web Content
6046 kworker/6:1-mm_percpu_wq
6048 kworker/18:2-events
6095 kworker/4:2-events_power_efficient
6240 kworker/5:2
6247 kworker/u64:0-kvfree_rcu_reclaim
6307 kworker/2:0-mm_percpu_wq
6318 ps
user@debian:~$
```



2026. "Año del Humanismo Mexicano en el Estado de México"

6. Ver procesos ordenados por uso de CPU (ps -eo pid,comm,%cpu --sort=-%cpu)

Este comando permite mostrar los procesos del sistema junto con su PID, nombre y porcentaje de uso de CPU, ordenándolos de mayor a menor consumo.

```
user@debian:~$ ps -eo pid,comm,%cpu --sort=-%cpu
PID COMMAND %CPU
2998 Isolated Web Co 21.0
2692 firefox-esr 12.3
1851 gnome-shell 9.6
5561 gnome-terminal 4.0
348 irq/183-ASCE120 1.1
3004 Isolated Web Co 0.8
2151 gnome-software 0.5
855 kworker/u6:1-i 0.2
1595 wpa_supplicant 0.2
2878 Privileged Cont 0.2
1985 ibus-daemon 0.1
6364 Web Content 0.1
177 kworker/u6:0-i 0.1
1 systemd 0.1
2460 ibus-extension 0.1
6517 kworker/7:2-191 0.1
2542 fwupd 0.0
6022 Web Content 0.0
979 irq/285-rtwap_p 0.0
5674 Web Content 0.0
1541 dbus-daemon 0.0
1714 pipewire 0.0
1594 NetworkManager 0.0
363 kworker/7:1-191 0.0
1717 wireplumber 0.0
2469 localsearch-3 0.0
18 rcu_preempt 0.0
2481 ibus-engine-sim 0.0
371 usb-storage 0.0
1712 dbus-daemon 0.0
768 kworker/u6:4-6-k 0.0
1545 polkitd 0.0
2836 WebExtensions 0.0
3127 Isolated Web Co 0.0
334 kworker/7:1H-ev 0.0
1718 pipewire-pulse 0.0
2154 evolution-alarm 0.0
784 kworker/u6:14- 0.0
2017 evolution-sourc 0.0
6247 kworker/u6:4-8-e 0.0
2254 Xwayland 0.0
56 migration/12 0.0
1690 systemd 0.0
5716 kworker/4:1-mm 0.0
2325 xdg-desktop-por 0.0
2844 RDP Process 0.0
3611 gnome-calendar 0.0
62 migration/13 0.0
2470 xdg-desktop-por 0.0
587 systemd-journal 0.0
2088 gsd-sharing 0.0
1998 gsd-media-keys 0.0
1549 systemd-logind 0.0
68 migration/14 0.0
74 migration/15 0.0
2385 evolution-adre 0.0
2015 gsd-wacom 0.0
1986 upowerd 0.0
120 khugepaged 0.0
2981 Utility Process 0.0
1550 udiskd 0.0
136 kworker/2:1-mm 0.0
11 kworker/0:1-mm 0.0
5915 kworker/8:0-mm 0.0
112 kworker/11:0-ev 0.0
1920 colord 0.0
645 systemd-udev 0.0
2409 mutter-x11-frm 0.0
1536 avahi-daemon 0.0
1994 gsd-keyboard 0.0
1999 gsd-power 0.0
1989 gsd-color 0.0
356 kworker/10:1-mm 0.0
356 kworker/6:2-eve 0.0
1815 gnome-session-b 0.0
1629 ModemManager 0.0
132 kworker/12:1-mm 0.0
5871 kworker/15:1-mm 0.0
2329 evolution-calen 0.0
336 kworker/8:1H-kb 0.0
2773 Socket Process 0.0
2240 gvfs-vfs-volume 0.0
1720 gnome-keyring-d 0.0
847 kworker/15:2-ev 0.0
6431 kworker/10:0-mm 0.0
2515 xdg-desktop-por 0.0
2359 gsd-xsettings 0.0
32 migration/4 0.0
38 migration/6 0.0
58 migration/10 0.0
6848 kworker/10:2-ev 0.0
1537 bluetoothd 0.0
26 migration/2 0.0
44 migration/8 0.0
5650 kworker/13:0-mm 0.0
1738 gnome-session-b 0.0
1546 smartd 0.0
357 kworker/9:1-191 0.0
338 kworker/10:1H-k 0.0
135 kworker/1:1-mm 0.0
4316 kworker/8:1-cgr 0.0
5568 bash 0.0
2622 power-profiles 0.0
2333 xdg-document-po 0.0
2070 gjs 0.0
1991 gsd-housekeepin 0.0
1954 gnome-shell-cal 0.0
1662 cupsd 0.0
1535 accounts-daemon 0.0
126 kcompactd0 0.0
21 migration/8 0.0
2018 gvfs-udisks2-vo 0.0
1804 gvfsd 0.0
340 kworker/5:1-eve 0.0
292 kworker/9:1H-ev 0.0
80 migration/1 0.0
86 migration/3 0.0
92 migration/5 0.0
98 migration/7 0.0
104 migration/9 0.0
110 migration/11 0.0
17 ksoftirqd/0 0.0
```



2026. "Año del Humanismo Mexicano en el Estado de México"

Oct 4 3:26 PM		
user@debian: ~		
70	kworrr/14:0-ev	0.0
4539	kworrr/14:2-ev	0.0
2467	ibus-portal	0.0
1964	gjs	0.0
1716	pipewire	0.0
1682	gdm-session-wor	0.0
1630	boltd	0.0
330	kworrr/11:1H-k	0.0
331	kworrr/6:1H-kb	0.0
170	kworrr/12:1H-k	0.0
27	kssoftirqd/2	0.0
51	kssoftirqd/10	0.0
3668	gvfsd-trash	0.0
2238	gss-identity-se	0.0
2089	gss-smartcard	0.0
1869	dbus-daemon	0.0
1730	rslit-daemon	0.0
1721	april-proxy	0.0
1675	gdm3	0.0
1547	switcheroo-cont	0.0
1544	low-memory-mon	0.0
266	kworrr/13:1H-e	0.0
99	kssoftirqd/7	0.0
33	kssoftirqd/4	0.0
39	kssoftirqd/6	0.0
45	kssoftirqd/8	0.0
4169	kworrr/3:1-eve	0.0
5804	kworrr/3:0-eve	0.0
2229	gss-daemon	0.0
2218	gvfs-gphoto2-vo	0.0
2813	gss-sound	0.0
1850	at-spi-bus-laun	0.0
275	kworrr/2:1H-ev	0.0
290	kworrr/4:1H-ev	0.0
243	kworrr/8:1H-ev	0.0
166	hwrng	0.0
81	kssoftirqd/1	0.0
57	kssoftirqd/12	0.0
63	kssoftirqd/13	0.0
2508	ibus-x11	0.0
2459	ibus-efconf	0.0
2446	cups-browsed	0.0
2387	gvfsd-metadata	0.0
2224	gvfs-goa-volume	0.0
2115	gvfs-ntp-volume	0.0
2080	gss-print-notif	0.0
2080	gss-rfkill	0.0
2086	gss-screensaver	0.0
2014	gss-usb-protect	0.0
1987	gss-ally-settin	0.0
1990	gss-datetime	0.0
1962	dconf-service	0.0
1919	at-spi2-registr	0.0
1894	xdp-permission	0.0
1794	sah-agent	0.0
1539	cron	0.0
333	kworrr/15:1H-e	0.0
335	kworrr/3:1H-ev	0.0
337	kworrr/5:1H-ev	0.0
93	kssoftirqd/5	0.0
105	kssoftirqd/9	0.0
111	kssoftirqd/11	0.0
-- -- -- -- --		
Oct 4 3:26 PM		
user@debian: ~		
69	kssoftirqd/14	0.0
75	kssoftirqd/15	0.0
2	kthreadd	0.0
3	pool_workqueue	0.0
4	kworrr/R-kvfre	0.0
5	kworrr/R-rcu_g	0.0
6	kworrr/R-sync	0.0
7	kworrr/R-slab	0.0
8	kworrr/R-netns	0.0
10	kworrr/8:0H-ev	0.0
13	kworrr/R-mm-pe	0.0
14	rcu_tasks_kthre	0.0
15	rcu_tasks_rude	0.0
16	rcu_tasks_trace	0.0
19	rcu_exp_gp_kthr	0.0
20	rcu_exp_gp_kthr	0.0
22	idle_inject/8	0.0
23	cpuhp/8	0.0
24	cpuhp/2	0.0
25	idle_inject/2	0.0
29	kworrr/2:0H-ev	0.0
30	cpuhp/4	0.0
31	idle_inject/4	0.0
35	kworrr/4:0H-ev	0.0
36	cpuhp/6	0.0
37	idle_inject/6	0.0
41	kworrr/6:0H-ev	0.0
42	cpuhp/8	0.0
43	idle_inject/8	0.0
47	kworrr/8:0H-ev	0.0
48	cpuhp/10	0.0
49	idle_inject/10	0.0
53	kworrr/10:0H-e	0.0
54	cpuhp/12	0.0
55	idle_inject/12	0.0
59	kworrr/12:0H-e	0.0
60	cpuhp/13	0.0
61	idle_inject/13	0.0
65	kworrr/13:0H-e	0.0
66	cpuhp/14	0.0
67	idle_inject/14	0.0
71	kworrr/14:0H-ev	0.0
72	cpuhp/15	0.0
73	idle_inject/15	0.0
77	kworrr/15:0H-e	0.0
78	cpuhp/1	0.0
79	idle_inject/1	0.0
83	kworrr/1:0H-ev	0.0
84	cpuhp/3	0.0
85	idle_inject/3	0.0
87	kssoftirqd/3	0.0
89	kworrr/3:0H-ev	0.0
90	cpuhp/5	0.0
91	idle_inject/5	0.0
95	kworrr/5:0H-ev	0.0
96	cpuhp/7	0.0
97	idle_inject/7	0.0
101	kworrr/7:0H-ev	0.0
102	cpuhp/9	0.0
103	idle_inject/9	0.0
107	kworrr/9:0H-ev	0.0
108	cpuhp/11	0.0



2026. "Año del Humanismo Mexicano en el Estado de México"

```
Oct 4 3:27 PM
user@debian:~$ ps
169  idle_inject/11 0.0
113 kworker/11:0H-e 0.0
120 kdevfsd 0.0
121 kworker/R-inet 0.0
122 kauditd 0.0
123 khungtaskd 0.0
124 oom_reaper 0.0
125 kworker/R-write 0.0
127 ksmd 0.0
129 kworker/R-kint 0.0
138 kworker/R-kllac 0.0
131 kworker/R-blkc 0.0
133 irq/9-acpi 0.0
134 kworker/13:1-cg 0.0
139 kworker/R-tpm_d 0.0
148 kworker/R-edac 0.0
141 kworker/R-devfr 0.0
142 kworker/R-quota 0.0
145 kworker/14:1H-e 0.0
146 kswapd0 0.0
154 kworker/R-kthro 0.0
158 irq/122-aerdrv 0.0
159 irq/123-aerdrv 0.0
160 irq/123-pciehp 0.0
161 irq/124-aerdrv 0.0
162 irq/125-aerdrv 0.0
163 irq/127-pciehp 0.0
164 irq/129-pciehp 0.0
165 kworker/R-acpi 0.0
167 kworker/R-hfi_u 0.0
168 kworker/R-mlid 0.0
169 kworker/R-ipwk 0.0
175 kworker/R-kstrp 0.0
332 kworker/1:1H-ev 0.0
343 watchdogd 0.0
347 irq/166-ITE5570 0.0
350 irq/184-aerdrv 0.0
352 kworker/R-nvme 0.0
353 kworker/R-nvme 0.0
354 kworker/R-nvme 0.0
355 kworker/R-nvme 0.0
358 kworker/R-ttm 0.0
359 card0-crtc0 0.0
368 card0-crtc1 0.0
361 card0-crtc2 0.0
362 card0-crtc3 0.0
368 kworker/R-USBC0 0.0
369 scsi_eh_0 0.0
370 kworker/R-scsi 0.0
372 kworker/R-uas 0.0
381 kworker/R-md 0.0
382 kworker/R-md_b1 0.0
388 kworker/R-raid5 0.0
573 psimon 0.0
647 psimon 0.0
737 irq/283-mei_me 0.0
743 kworker/R-crypt 0.0
744 kworker/R-led_w 0.0
749 kworker/R-clg0 0.0
750 irq/16-processo 0.0
1551 avahi-daemon 0.0
1681 psimon 0.0

1646 krfcommd 0.0
1692 (sd-pan) 0.0
1734 gdm-wayland-se 0.0
1792 gcr-ssh-agent 0.0
1793 gnome-session-c 0.0
1812 gvfsd-fuse 0.0
2143 gsd-disk-utilit 0.0
2382 gsd-printer 0.0
2342 fusermount3 0.0
2700 crashhelper 0.0
4166 kworker/9:2-eve 0.0
4343 kworker/1:0 0.0
4551 kworker/11:1 0.0
5725 kworker/7:0-eve 0.0
5731 kworker/2:2-eve 0.0
5796 kworker/12:2-mm 0.0
5813 kworker/5:0-eve 0.0
5826 kworker/0:0-eve 0.0
6846 kworker/6:1-mm 0.0
6895 kworker/4:2-eve 0.0
6248 kworker/3:2-eve 0.0
6307 kworker/2:0-mm 0.0
6332 kworker/8:2-mm 0.0
6417 kworker/0:2 0.0
6419 kworker/15:0-mm 0.0
6448 kworker/4:0-eve 0.0
6449 kworker/3:2-mm 0.0
6527 kworker/9:0 0.0
6532 ps 0.0

user@debian:~$ ^C
user@debian:~$
```

Conclusión

En esta práctica se logró conocer y utilizar el comando ps en Linux para consultar los procesos que se encuentran activos en el sistema. Mediante sus diferentes opciones fue posible visualizar los procesos del usuario actual, obtener información más detallada, filtrar procesos específicos y consultar los procesos asociados a un usuario determinado. También se aprendió a mostrar el PID y nombre de los procesos, así como a ordenarlos de acuerdo con su consumo de CPU. En general, la práctica permitió comprender mejor cómo Linux administra sus procesos y cómo la terminal puede utilizarse para obtener información útil sobre el funcionamiento del sistema.